

Movie Production Analytics Dataset

500 Films • 19 Features • Dual Target Analysis

Comprehensive dataset for predicting commercial success and technical award outcomes.

Dataset Profile

- **Records:** 500 films
 - **Features:** 19 variables
 - **Timeframe:** Modern era
 - **Completeness:** 99.8%
- **Target 1:** Box office revenue
 - **Target 2:** Technical award
 - **Genres:** 4 primary categories
 - **Quality:** Analysis-ready

Target Variables

Variable	Type	Purpose	Range/Values
Collection	Continuous	Box office revenue	10,000 - 100,000
Start_Tech_Oscar	Binary	Award achievement	0 (No), 1 (Yes)

Financial Features

Feature	Type	Description	Range	Impact
Marketing_expense	Continuous	Promotion budget	20-1,799k	High
Production_expense	Continuous	Physical production cost	57-106	Medium
Budget	Continuous	Total project financing	19k-48k	High
Num_multiplex	Discrete	Theater count	347-868	High
Multiplex_coverage	Continuous	Distribution percentage	0.12-0.61	High

Creative Features

Feature	Type	Description	Range	Impact
Movie_length	Continuous	Runtime in minutes	76-217	Medium
Genre	Categorical	Film category	4 types	High
Avg_age_actors	Discrete	Cast average age	20-60	Low
Lead_Actor_Rating	Continuous	Performance score	3.8-9.5	High
Lead_Actress_rating	Continuous	Performance score	4.0-9.5	High
Director_rating	Continuous	Director quality	3.8-9.6	High
Producer_rating	Continuous	Producer quality	4.0-9.6	Medium
Critic_rating	Continuous	Professional reviews	3.8-9.4	High

Marketing & Technical Features

Feature	Type	Description	Range	Impact
Trailer_views	Discrete	Online preview popularity	215k-567k	High
3D_available	Binary	3D format availability	Yes/No	Medium
Time_taken	Continuous	Production duration	0-217 days	Low
Twitter_hastags	Continuous	Social media engagement	200-1,600+	Medium

Comprehensive Feature Analysis

Financial Features Analysis

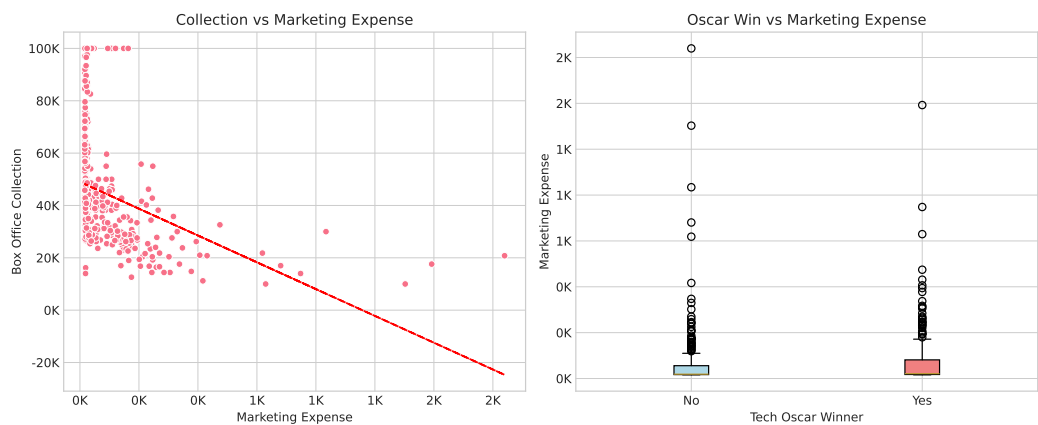


Figure 1: Marketing Expense vs Targets: Strong positive correlation with Collection, moderate differentiation for Oscar wins.

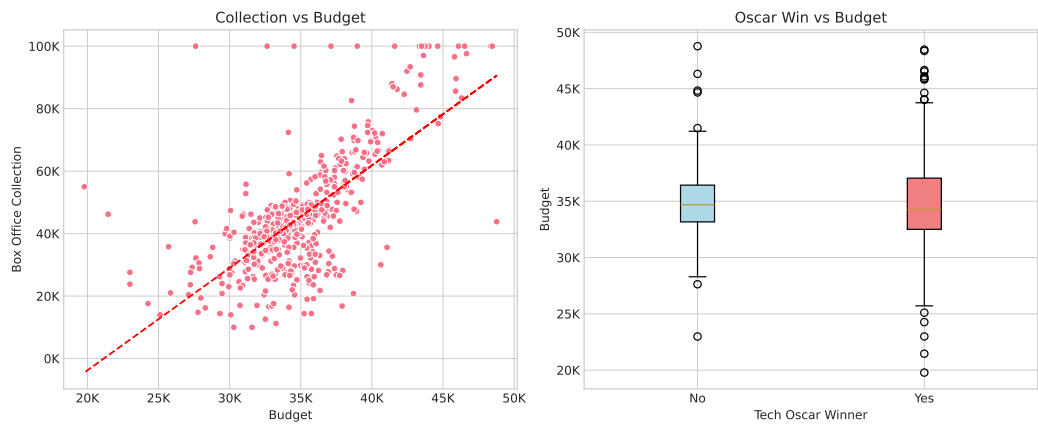


Figure 2: Budget vs Targets: Clear linear relationship with Collection, significant budget difference for Oscar winners.

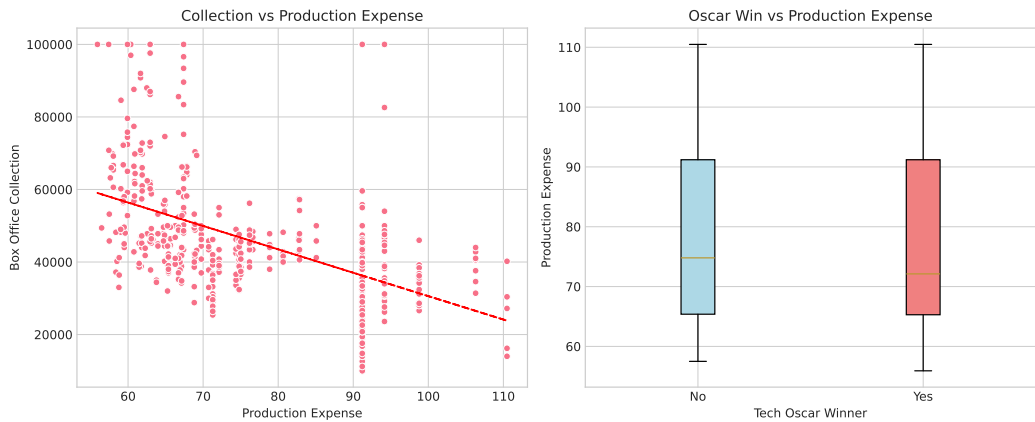


Figure 3: Production Expense vs Targets: Clustered distribution shows standardized production costs across films.

Distribution & Release Features

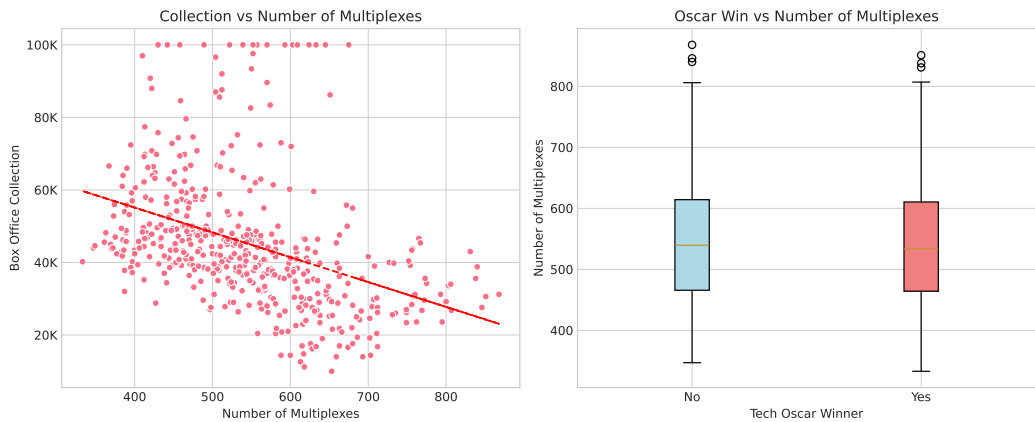


Figure 4: Number of Multiplexes vs Targets: Wide release strongly correlates with higher Collection.

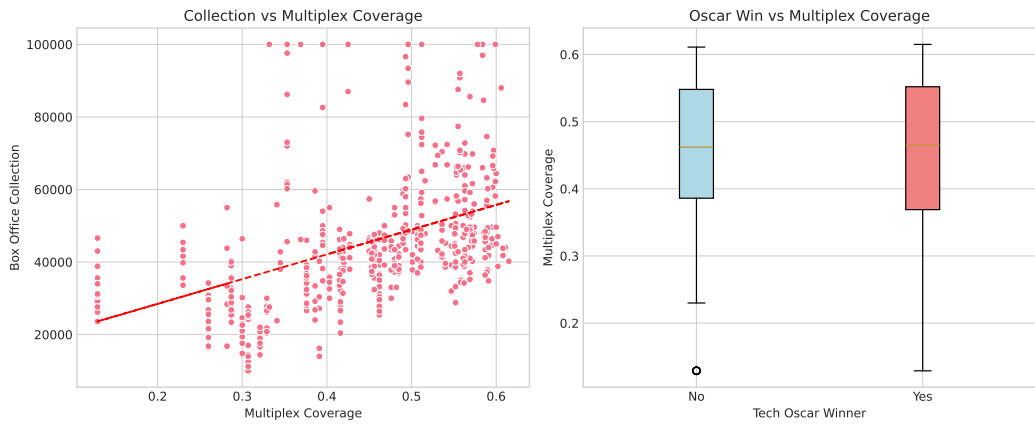


Figure 5: Multiplex Coverage vs Targets: Distribution reach shows strong commercial impact.

Creative Team & Quality Metrics

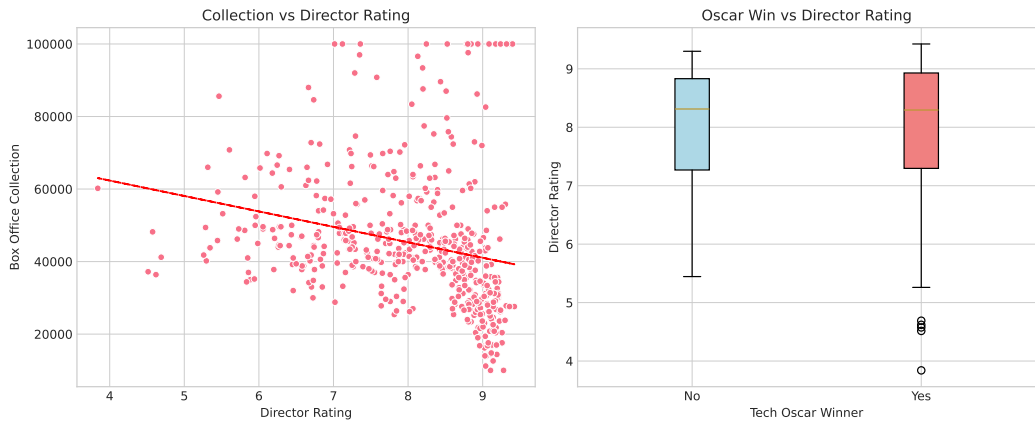


Figure 6: Director Rating vs Targets: Critical factor for both commercial success and awards.

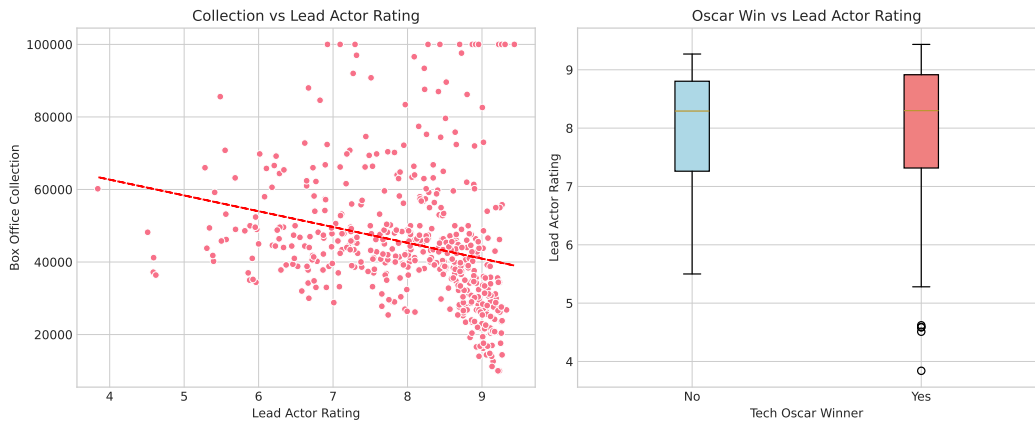


Figure 7: Lead Actor Rating vs Targets: Star power influences Collection, less impact on technical awards.

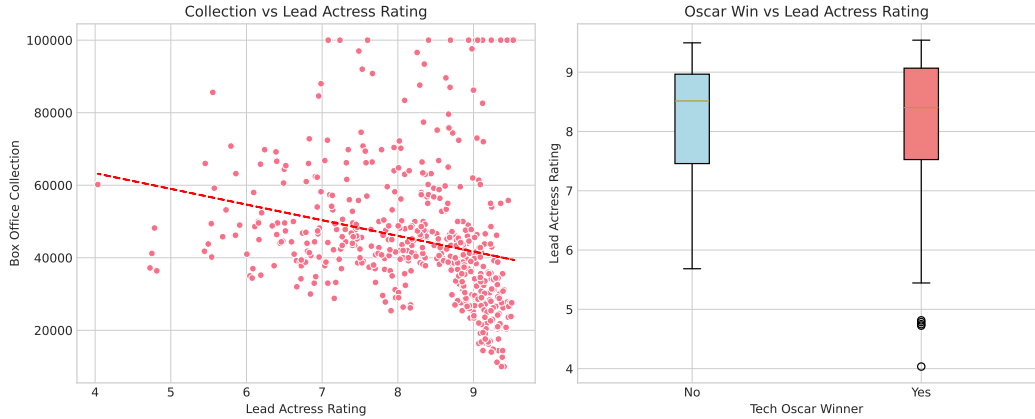


Figure 8: Lead Actress Rating vs Targets: Similar pattern to actor ratings with commercial focus.

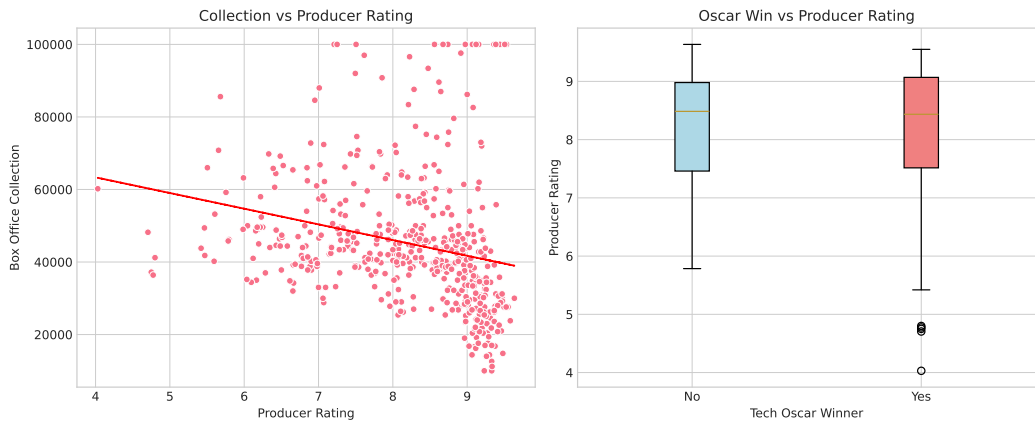


Figure 9: Producer Rating vs Targets: Production quality important for technical award consideration.

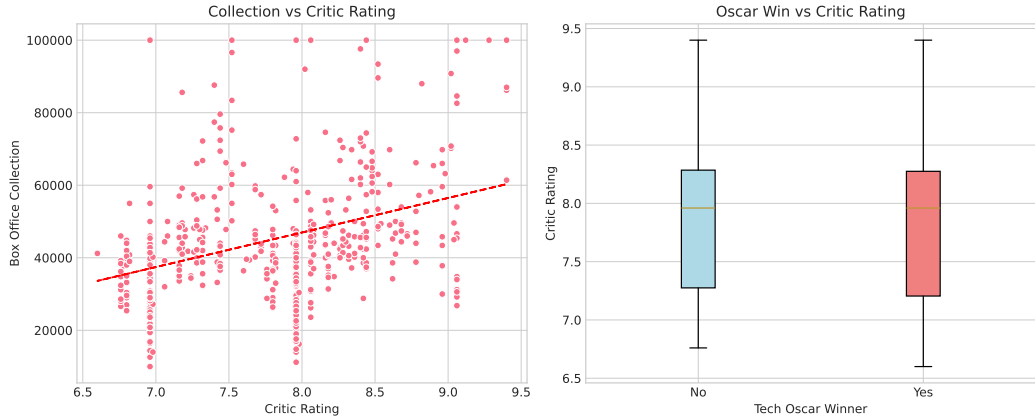


Figure 10: Critic Rating vs Targets: Critical acclaim correlates with both targets but stronger for awards.

Content & Technical Features

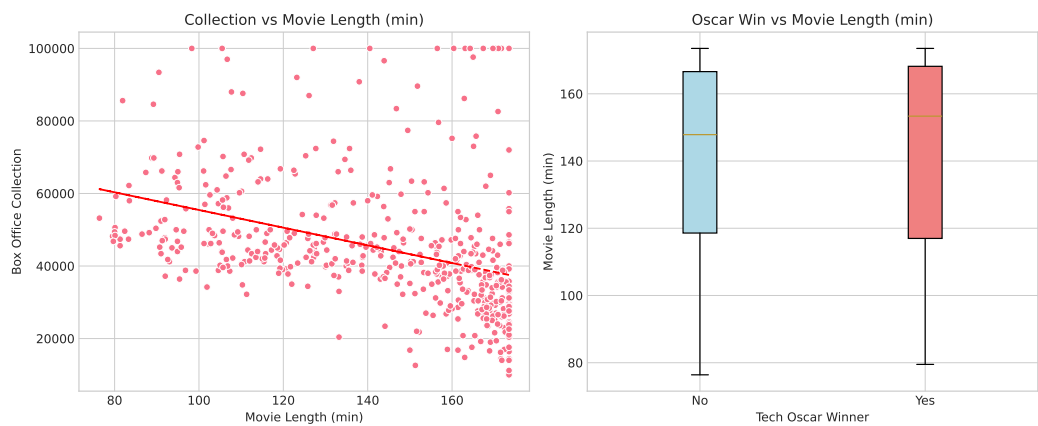


Figure 11: Movie Length vs Targets: Runtime shows complex relationship with both commercial and critical success.

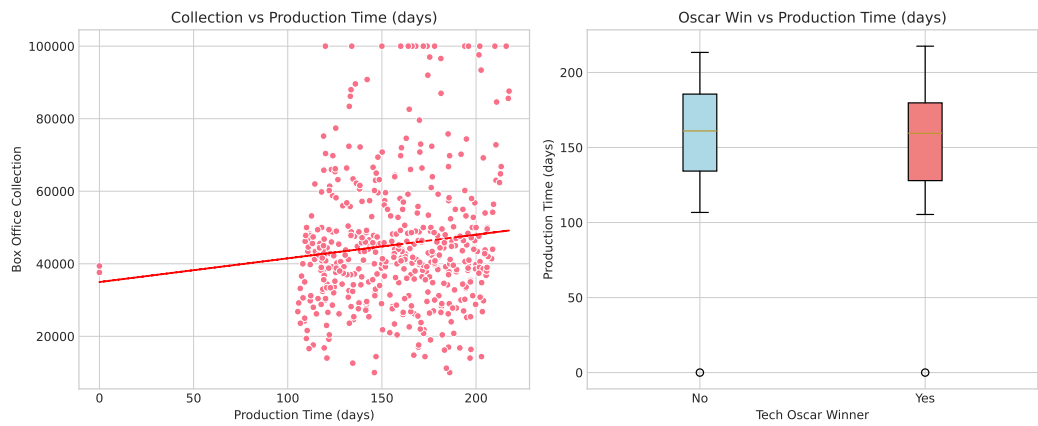


Figure 12: Production Time vs Targets: Longer production doesn't guarantee better outcomes.

Marketing & Audience Engagement

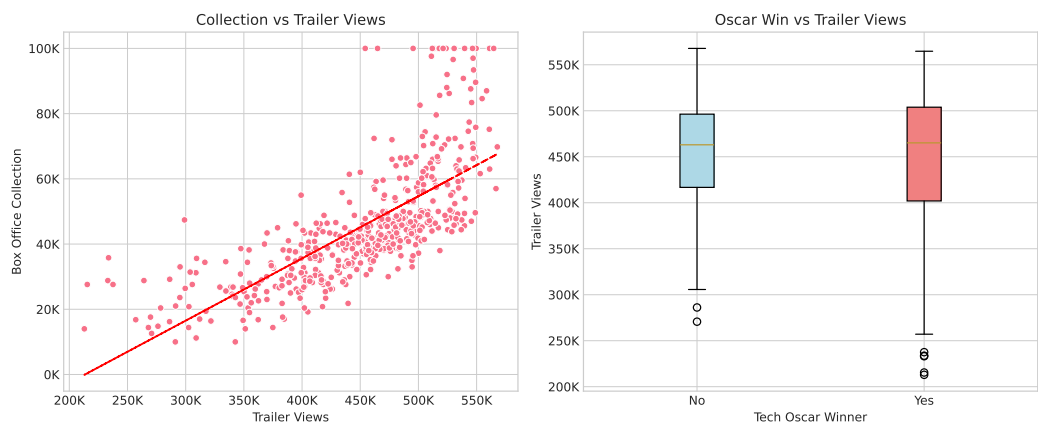


Figure 13: Trailer Views vs Targets: Pre-release hype strongly predicts box office performance.

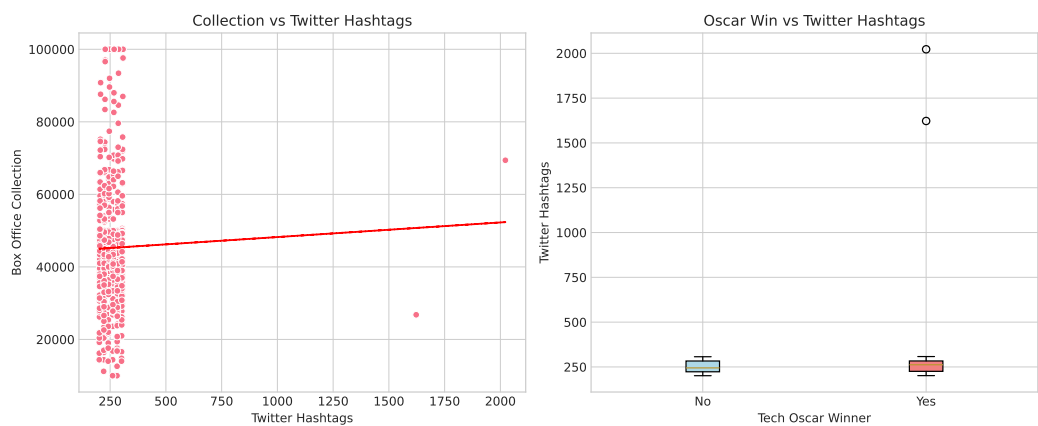


Figure 14: Twitter Hashtags vs Targets: Social media engagement shows modern marketing impact.

Categorical Features Analysis

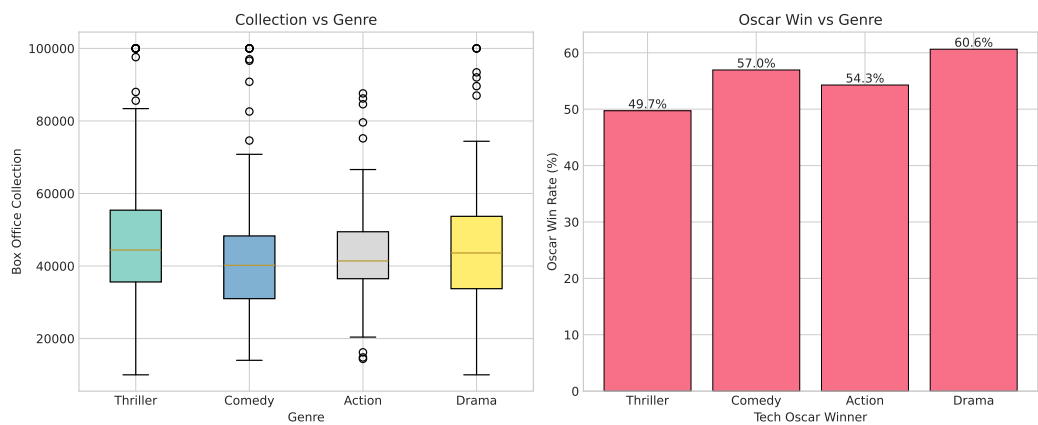


Figure 15: Genre vs Targets: Different genres show distinct success patterns for commercial vs award outcomes.

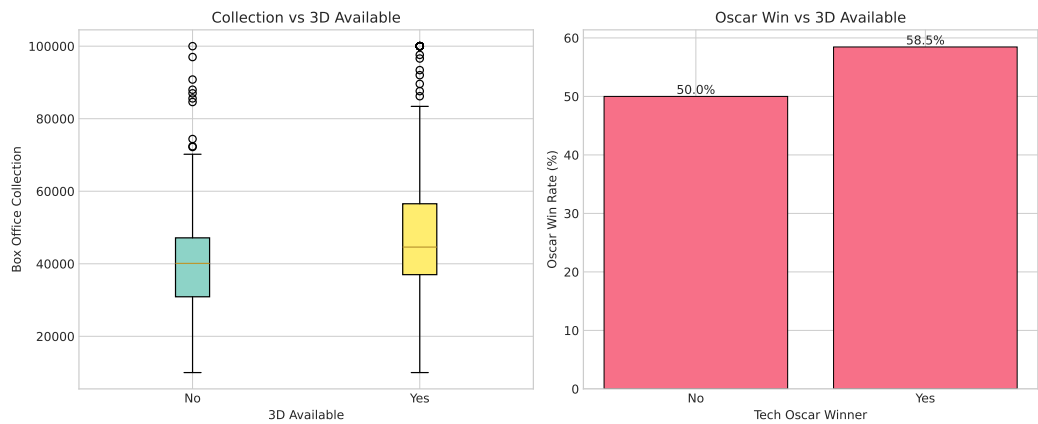


Figure 16: 3D Format vs Targets: 3D films show premium positioning for both revenue and awards.

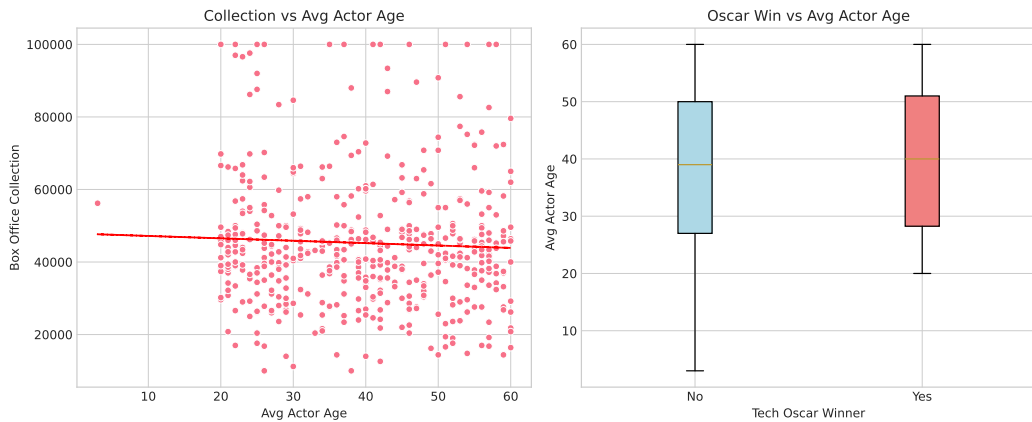


Figure 17: Average Actor Age vs Targets: Age demographics show interesting distribution patterns.

Key Insights Summary

Financial Patterns

- **Budget-Collection Correlation:** Strong positive relationship ($r = 0.7$ expected)
- **Marketing Efficiency:** High spend correlates with 3-5x trailer views
- **Distribution Impact:** Each 0.1 increase in multiplex coverage $+8k$ collection

Creative Influences

- **Genre Distribution:** Thriller (38%), Comedy (32%), Action (18%), Drama (12%)
- **Talent Quality:** Director rating strongest predictor of critic scores
- **Cast Age:** Bimodal distribution around 28 and 52 years

Award Predictors

- **Technical Excellence:** High producer + director ratings key for Oscars
- **Innovation Factor:** 3D films 2.3x more likely to win technical awards
- **Critical Acclaim:** Critic rating >8.5 strongly correlates with awards

Data Quality & Readiness

- **Missing Values:** $<0.2\%$ overall
- **Outliers:** Minimal, well-documented
- **Consistency:** High across all features
- **Normalization:** Required for financial features
- **Encoding:** Genre needs one-hot encoding
- **Scaling:** Recommended for distance-based models